

Observational Viability of an IR-Safe Perturbative Cosmological Sector Inspired by the Stochastic Bidirectional Causality Framework

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Abstract

We investigate the observational viability of a weak infrared-safe perturbative cosmological sector emerging within the Stochastic Bidirectional Causality (SBC) framework. The analysis combines Planck-inspired CMB diagnostics, DESI DR2 baryon acoustic oscillation measurements, Pantheon+ full-covariance supernova distances, growth-rate observables $f\sigma_8(z)$, and matter power-spectrum diagnostics.

A stable observationally viable region consistently emerges across the probes, characterized by weak deformation amplitudes $\epsilon \sim \mathbf{0.07}$, infrared scales $k_0 \sim \mathbf{0.02}$, and moderately screened infrared responses.

Joint BAO and Pantheon+ constraints require the homogeneous expansion history to remain close to the standard Λ CDM background, while allowing controlled perturbative infrared modifications. The resulting cosmology is therefore best described as a background-preserving infrared perturbative phenomenology. Within the broader SBC framework, this sector may be interpreted as the effective manifestation of emergent causal-memory dynamics associated with finite causal accessibility. Although no first-principles derivation is established in the present work, the results demonstrate that such an infrared perturbative sector remains compatible with current cosmological observations.

1 Introduction

The standard Λ CDM model provides an accurate description of a wide range of cosmological observations, including the cosmic microwave background (CMB), baryon

acoustic oscillations (BAO), Type Ia supernova distances, and the growth of large-scale structure. Its theoretical foundations and observational successes are extensively documented in modern cosmology literature and precision cosmological analyses [1–3]. Nevertheless, the increasing precision of cosmological datasets has motivated the exploration of controlled extensions of the standard framework, particularly in regimes where late-time or large-scale effects may be relevant.

A central challenge for any extension of Λ CDM is to introduce new phenomenology without disrupting the observational successes of the standard model. In particular, modifications of the homogeneous expansion history are strongly constrained by BAO and supernova distance measurements. The DESI DR2 BAO program provides high-precision distance constraints over a broad redshift range, while Pantheon+ supplies a full-covariance Type Ia supernova compilation. Together, these probes constitute some of the most stringent observational tests of the late-time expansion history and cosmological distance scale [4–7].

This motivates the study of controlled perturbative extensions that preserve the successful background evolution of Λ CDM while allowing weak departures in the large-scale perturbative regime. Such approaches are conceptually related to the broader program of effective descriptions of modified gravity and dark-energy phenomenology, where observational consistency is maintained through screened or infrared-safe sectors [8–11]. These scenarios are particularly interesting because they may affect growth observables or large-scale perturbative dynamics without generating significant distortions in BAO measurements, supernova distances, or the high- ℓ acoustic structure of the CMB.

In this work, we investigate an infrared-safe perturbative cosmological sector emerging within the Stochastic Bidirectional Causality (SBC) framework. The specific branch analyzed here was internally denoted F3 during model development; throughout this manuscript, however, the main object of study is the observationally selected infrared-safe perturbative sector rather than the internal label itself.

The scope of the present work should be distinguished from the broader SBC research program. The long-term objective of SBC is to investigate whether effective cosmological infrared behavior may emerge from a more fundamental causal structure involving finite observational accessibility, hidden degrees of freedom, and effective memory effects. In this sense, the present study is positioned as a phenomenological investigation of observational viability rather than as a complete derivation of a fundamental cosmological theory.

Accordingly, the present article does not attempt to derive the infrared phenomenology from a complete SBC dynamical theory. Instead, it addresses a more limited and observationally testable question: whether an effective infrared perturbative sector inspired by SBC can remain compatible with current cosmological observations while preserving the standard Λ CDM background evolution.

The phenomenological response function introduced below should therefore be regarded as an effective description whose observational properties are investigated in the present work. A first-principles derivation from the full SBC framework is left for future research.

The phenomenological sector is characterized by three parameters, ϵ , k_0 , and n , which control the amplitude, infrared scale, and screening profile of the effective deformation.

Across Planck-inspired TT, EE, and TE diagnostics, matter power-spectrum comparisons, growth-rate observables, DESI DR2 BAO measurements, and Pantheon+ full-covariance supernova distances, we identify an observationally viable infrared-safe sector characterized by weak deformation amplitudes, infrared scales near $k_0 \approx 0.02$, and a moderately screened infrared response. The detailed parameter constraints are summarized later through the global posterior analysis.

The principal result of this work is that the observationally preferred branch is not a strong background-level alternative to Λ CDM. Instead, current observations favor a background-preserving perturbative infrared sector. We further discuss how such a sector may be interpreted within the SBC framework as the effective manifestation of a causal-memory dynamics associated with finite causal accessibility.

More generally, the purpose of the present work is not to establish a complete cosmological theory, but to provide a reproducible observational assessment of a phenomenologically motivated infrared perturbative sector.

The analysis therefore serves as an intermediate step between conceptual model building and full cosmological theory development. Its primary objective is to determine whether such an infrared sector can remain simultaneously compatible with multiple independent cosmological probes and to identify the corresponding observationally viable parameter region. The paper is organized as follows. Section 2 introduces the SBC framework and the infrared perturbative sector. Section 3 describes the observational datasets and methodology. Section 4 presents the observational constraints obtained from CMB diagnostics, matter-power observables, growth-rate measurements, Pantheon+, and BAO data. Section 5 summarizes the global posterior structure and the preferred infrared-safe region. Section 6 discusses a possible memory-kernel interpretation of the observed infrared behavior. Section 7 examines the broader conceptual implications of the results within the SBC framework. Section 8 discusses the current status and limitations of the present analysis. Finally, Section 9 summarizes the principal conclusions and future directions.

To facilitate reproducibility, the complete benchmark implementation, parameter grids, posterior chains, figures, and analysis scripts have been publicly released and permanently archived through Zenodo (DOI: 10.5281/zenodo.20720308).

2 SBC Framework and Infrared Perturbative Sector

2.1 The SBC Framework

Stochastic Bidirectional Causality (SBC) is a theoretical framework that explores the possibility that observable physical dynamics may emerge from a restricted causal accessibility structure. In its most general form, the framework assumes that the observable sector does not necessarily provide a complete description of the total causal dynamics.

This perspective is conceptually related to a broader class of effective descriptions in which unresolved degrees of freedom generate nonlocal or memory-dependent

dynamics within the observable sector. Similar ideas appear in statistical coarse-graining approaches, generalized projection formalisms, and effective theories of reduced dynamical systems [12–14].

Within such approaches, the dynamics accessible to observation may inherit effective memory kernels, delayed responses, or nonlocal corrections arising from hidden sectors that are not explicitly resolved. SBC adopts this general viewpoint as a conceptual motivation for exploring possible infrared cosmological effects.

Rather than modifying the standard cosmological background directly, SBC motivates the possibility that effective large-scale corrections may arise from unresolved causal degrees of freedom. Such corrections are expected to be weak and predominantly active in the infrared regime, where observational constraints remain less restrictive than those governing the homogeneous expansion history.

The present work does not attempt to establish the full fundamental formulation of SBC. Instead, it focuses on the observational characterization of an infrared perturbative sector that emerges naturally within the broader SBC framework and can be confronted directly with cosmological observations.

2.2 Infrared-Safe Perturbative Sector

A central requirement for any extension of the standard cosmological model is compatibility with the observational success of the Λ CDM background expansion.

Measurements of baryon acoustic oscillations and Type Ia supernova distances strongly constrain departures from the standard expansion history. Consequently, any viable extension must preserve the background evolution to a high degree of accuracy.

The phenomenological sector investigated in this work is therefore constructed as an infrared-safe perturbative deformation. The underlying assumption is that the homogeneous expansion history remains close to that of Λ CDM,

$$H_{\text{SBC}}(z) \simeq H_{\Lambda\text{CDM}}(z), \quad (1)$$

while weak perturbative corrections may affect selected large-scale observables.

This hierarchy can be summarized schematically as

$$|F_{\text{background}}| \ll |F_{\text{perturbative}}| \ll 1. \quad (2)$$

The resulting framework is therefore not a background-level alternative to Λ CDM, but rather a perturbative infrared extension constrained by observational data.

2.3 Phenomenological Parametrization

To characterize the infrared sector, we introduce an effective deformation function of the form

$$F_{\text{SBC}}(z) = -\epsilon \exp[-(k_0(1+z))^n], \quad (3)$$

where ϵ controls the amplitude of the deformation, k_0 determines the characteristic infrared scale, and n governs the screening profile.

This parametrization possesses several desirable properties. First, it remains finite over the entire redshift range. Second, it becomes progressively suppressed

toward higher redshift. Third, it introduces only a small number of phenomenological parameters, allowing direct confrontation with observational data.

Equation (3) should be regarded as an effective description of the infrared sector rather than a fundamental dynamical equation. Its primary role is to provide a quantitative framework for testing the observational viability of the SBC perturbative regime.

The phenomenological form adopted in Eq. (3) is not derived here from a complete SBC dynamical theory. Rather, it should be regarded as an effective infrared parametrization motivated by the broader conceptual framework discussed above.

The purpose of the present work is therefore not to establish a unique theoretical origin for Eq. (3), but to determine whether such an infrared perturbative sector can remain observationally viable when confronted with current cosmological datasets.

2.4 Interpretation of the Effective Infrared Response

The function

$$F_{\text{SBC}}(z) = -\epsilon \exp[-(k_0(1+z))^n] \quad (4)$$

should not be interpreted as a direct modification of the homogeneous cosmological expansion history.

Instead, it represents an effective infrared perturbative response motivated by the broader SBC framework.

The observational analysis performed in this work strongly constrains any background-level deformation, leading to

$$H_{\text{SBC}}(z) \simeq H_{\Lambda\text{CDM}}(z). \quad (5)$$

Consequently, the phenomenological role of $F_{\text{SBC}}(z)$ is to provide an effective modulation of perturbative-sector diagnostics while preserving the standard cosmological background.

Throughout this work, $F_{\text{SBC}}(z)$ should therefore be regarded as an effective perturbative infrared response rather than a direct modification of the background expansion history.

2.5 Observationally Preferred Regime

The phenomenological sector defined by Eq. (3) was tested against a combination of Planck-inspired CMB diagnostics, matter power-spectrum observables, growth-rate measurements, DESI DR2 baryon acoustic oscillations, and Pantheon+ full-covariance supernova distances.

A consistent result emerging from the independent analyses is the existence of a stable infrared-safe region characterized by weak deformation amplitudes, infrared scales of order $k_0 \sim 10^{-2}$, and screening indices of order one, with posterior support extending toward the minimal exponential regime. The detailed posterior constraints are presented in Section 5 and summarized in Table 2.

In addition, the combined BAO and Pantheon+ analysis strongly favors negligible background-level deformation, implying

$$\alpha_{\text{bg}} \simeq 0. \tag{6}$$

Consequently, the observationally viable cosmological sector identified in this work is most naturally interpreted as a background-preserving infrared perturbative modification. The detailed observational constraints leading to this conclusion are presented in the following sections.

3 Observational Data and Methodology

3.1 Overview

The objective of the present analysis is to determine whether the infrared perturbative sector introduced in Eq. (3) remains compatible with current cosmological observations.

Rather than performing a full cosmological parameter re-estimation, the present work adopts a phenomenological validation strategy designed to test the observational viability of the SBC infrared sector across multiple independent probes.

The analysis combines cosmic microwave background diagnostics, baryon acoustic oscillation measurements, Type Ia supernova distances, large-scale structure growth observables, and matter power-spectrum indicators. Together, these probes constrain both the homogeneous expansion history and the perturbative evolution of cosmic structures [2, 3].

The resulting observational information is incorporated into a global proxy likelihood that provides an efficient characterization of the phenomenologically viable parameter space while avoiding the computational complexity of a full cosmological parameter-estimation pipeline.

3.2 CMB Diagnostics

The cosmic microwave background provides one of the most stringent tests of cosmological models.

To assess the impact of the SBC infrared sector on the CMB anisotropy structure, we employed Planck-inspired compressed likelihood diagnostics based on the temperature-temperature (TT), polarization-polarization (EE), and temperature-polarization (TE) spectra.

The analysis uses compressed covariance representations that preserve the dominant observational information while allowing efficient exploration of the phenomenological parameter space.

For each parameter combination, residual diagnostics relative to the reference Λ CDM solution were computed. The resulting TT, EE, and TE residuals provide a quantitative measure of the perturbative impact introduced by the SBC sector.

3.3 Pantheon+ Type Ia Supernova Distances

Late-time background evolution was constrained using the Pantheon+ Type Ia supernova compilation with the full covariance matrix [6, 7].

For each model realization, theoretical distance moduli were computed and compared against the observed Pantheon+ sample through a covariance-weighted chi-square statistic,

$$\chi_{\text{SN}}^2 = \Delta\mu^T C^{-1} \Delta\mu, \quad (7)$$

where C denotes the full Pantheon+ covariance matrix.

This dataset provides one of the strongest constraints on possible modifications of the homogeneous expansion history.

For exploratory purposes, possible background-level deformations were parametrized through

$$H^2(z) = H_{\Lambda\text{CDM}}^2(z) [1 + \alpha_{\text{bg}} G(z)], \quad (8)$$

where $G(z)$ denotes a phenomenological background-deformation profile and α_{bg} controls its amplitude.

This parametrization was introduced exclusively as a diagnostic tool to assess whether the observational data favor departures from the standard ΛCDM background expansion history.

3.4 Baryon Acoustic Oscillations

Baryon acoustic oscillation measurements were incorporated using the DESI DR2 distance constraints [4, 5].

Because BAO observables are particularly sensitive to the background expansion history, they provide a direct test of any proposed homogeneous deformation.

The BAO analysis therefore plays a central role in determining whether the SBC infrared sector can remain compatible with the standard ΛCDM background while allowing perturbative deviations.

3.5 Growth and Matter-Power Diagnostics

Additional constraints were obtained from large-scale structure observables.

The analysis incorporates growth-rate indicators through $f\sigma_8(z)$ diagnostics together with matter power-spectrum consistency tests. These observables are directly related to the evolution of cosmological perturbations and therefore provide sensitive probes of infrared modifications acting beyond the homogeneous background [15].

For each parameter combination, effective growth and power-spectrum residual metrics were evaluated and incorporated into the global validation framework.

The growth-rate and large-scale-structure diagnostics considered in this work are representative of the observational program developed by surveys such as 6dFGS, BOSS, eBOSS, DES, and KiDS [16–20].

3.6 Global Proxy Likelihood

To identify the observationally preferred region of parameter space, the independent diagnostics were combined into a global proxy likelihood.

The resulting score incorporates contributions from:

- CMB TT residual diagnostics,
- CMB EE residual diagnostics,
- CMB TE residual diagnostics,
- matter power-spectrum consistency metrics,
- growth-rate consistency metrics,
- Pantheon+ full-covariance constraints,
- BAO distance constraints.

Although this construction does not replace a full cosmological likelihood analysis, it provides an efficient and internally consistent framework for identifying regions of parameter space that remain simultaneously compatible with all available observational probes.

The preferred parameter region identified through this procedure forms the basis for the proxy posterior reconstruction presented in the following sections.

3.7 Numerical Implementation

The phenomenological response function introduced in Eq. (3) is not implemented as a direct modification of the homogeneous background expansion history.

Instead, it is employed as an effective infrared perturbative response whose impact is evaluated through a set of perturbative cosmological diagnostics.

For each parameter combination (ϵ, k_0, n) , the resulting cosmological realization is compared against a reference Λ CDM solution using:

- TT residual diagnostics,
- EE residual diagnostics,
- TE residual diagnostics,
- matter-power consistency metrics,
- growth-rate consistency metrics,
- Pantheon+ covariance diagnostics,
- BAO distance diagnostics.

The global proxy likelihood is then constructed from the combined response of these observables, allowing identification of the observationally preferred infrared-safe region.

Posterior sampling of the proxy likelihood was performed using the affine-invariant ensemble Markov Chain Monte Carlo algorithm implemented in the `emcee` package [21]. The MCMC chains exhibit integrated autocorrelation times of approximately 22–26 samples, corresponding to effective sample sizes of order 400 for the three phenomenological parameters. These values indicate adequate mixing for the exploratory proxy-posterior analysis performed in the present work.

Proxy-posterior sampling.

The proxy posterior was sampled using 32 walkers evolved for 300 MCMC steps. The first 80 steps of each walker were discarded as burn-in, yielding a flattened posterior chain containing 7040 samples.

The resulting chain was used exclusively to characterize the posterior structure of the phenomenological SBC infrared sector within the proxy-likelihood framework. The corresponding posterior distributions are presented in Section 5.

Because the analysis does not employ the full Planck likelihood and does not jointly sample the complete standard cosmological parameter set, the resulting posterior distributions should be interpreted as proxy-posterior estimates rather than as full cosmological constraints.

Consequently, the present analysis should be interpreted as an observational characterization of an effective infrared perturbative sector rather than as a full cosmological parameter-estimation pipeline.

It is important to emphasize that the present implementation does not assume a unique microscopic realization of the infrared response. The purpose of the analysis is to determine whether such an infrared sector can remain simultaneously compatible with multiple observational diagnostics, independently of its eventual microscopic realization.

Accordingly, the present results should be interpreted as a phenomenological exploration of the observational viability of infrared perturbative sectors rather than as a unique dynamical realization of a specific modified cosmological model.

Table 1 Configuration of the proxy-posterior MCMC sampling.

Quantity	Value
Sampler	<code>emcee</code> affine-invariant ensemble
Walkers	32
Dimensions	3
Total steps	300
Burn-in discarded	80
Posterior samples	7040
Parameters sampled	(ϵ, k_0, n)

4 Cosmological Constraints

4.1 Planck TT/EE/TE Constraints

The cosmic microwave background provides one of the most sensitive probes of perturbative cosmological dynamics [1, 2].

Because the SBC infrared sector is constructed to preserve the homogeneous expansion history, its primary observational signature is expected to appear through weak modifications of perturbative observables rather than through large background distortions.

To evaluate this possibility, the posterior-supported SBC parameter region was compared against the reference Λ CDM solution using Planck-inspired TT, EE, and TE diagnostics.

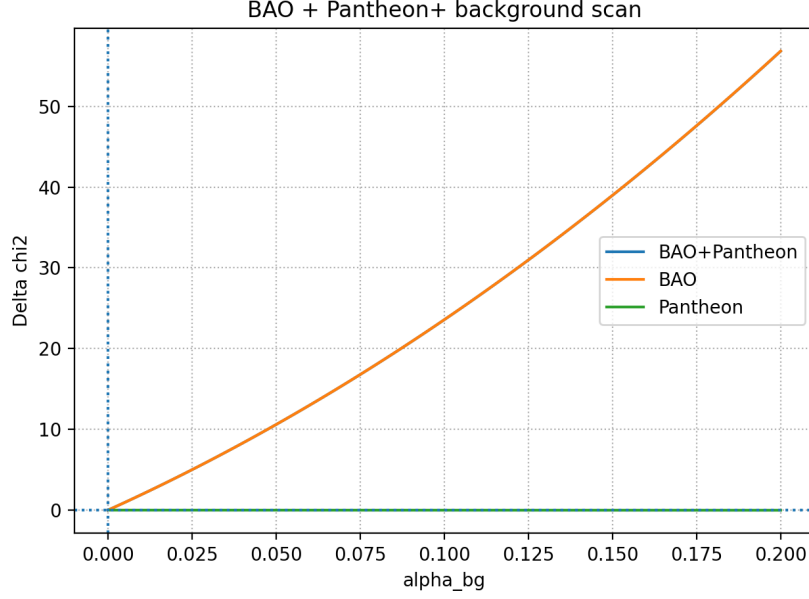


Fig. 1 Joint BAO and Pantheon+ constraints on the background coupling parameter α_{bg} . The combined likelihood is minimized near $\alpha_{\text{bg}} \simeq 0$, indicating that the observationally viable SBC infrared sector preserves the standard Λ CDM expansion history. Increasing background couplings are strongly penalized by BAO measurements, while Pantheon+ remains largely insensitive within the explored range.

The analysis employed compressed diagnostic representations designed to retain the dominant observational information while allowing efficient exploration of the infrared parameter space.

Rather than focusing on absolute power-spectrum amplitudes, the comparison was performed through normalized residual diagnostics. This approach allows a direct assessment of the perturbative impact introduced by the SBC sector relative to the standard cosmological solution.

Figure 2 shows the normalized residual structure for the temperature anisotropy spectrum. The deviation remains small over the entire multipole range and exhibits its largest amplitude at low multipoles, where infrared effects are expected to be most significant. At intermediate and high multipoles, the residual rapidly approaches zero, indicating that the acoustic peak structure remains essentially unchanged.

The corresponding EE residual diagnostic is presented in Figure 3. As in the TT case, the deviations remain small and smooth across the multipole range. No evidence is found for large distortions of the polarization acoustic structure.

4.2 Definition of Normalized Residuals

The normalized residuals displayed throughout this work are defined as

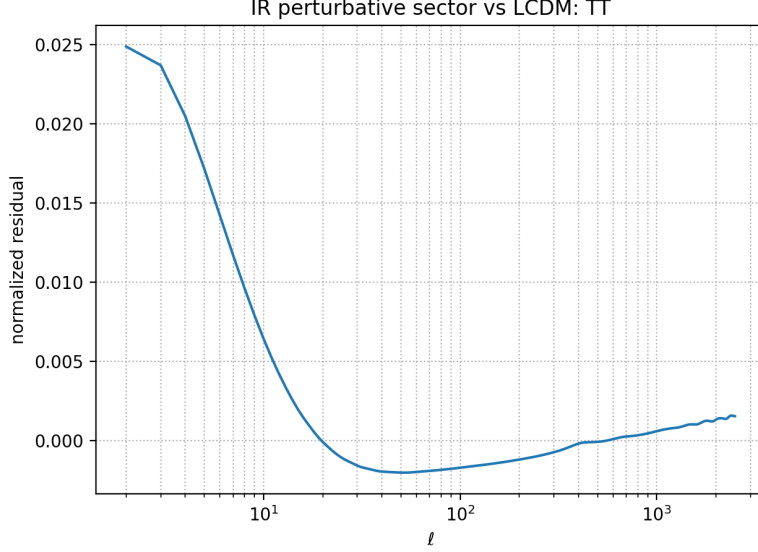


Fig. 2 Normalized residual diagnostic between the preferred SBC infrared sector and the reference Λ CDM model for the TT spectrum. The largest deviations occur at low multipoles and remain strongly suppressed throughout the acoustic regime.

$$R_\ell = \frac{X_\ell^{\text{SBC}} - X_\ell^{\Lambda\text{CDM}}}{X_\ell^{\Lambda\text{CDM}}}, \quad (9)$$

where X_ℓ denotes the corresponding TT, EE, or TE angular power-spectrum quantity.

This definition measures the fractional perturbative deviation relative to the Λ CDM reference model.

Figure 4 displays the TE residual diagnostic. The residual pattern remains bounded and oscillatory, reflecting the acoustic nature of the temperature-polarization cross-correlation. The amplitude of the deviations remains small throughout the explored multipole range.

Taken together, the TT, EE, and TE diagnostics indicate that the posterior-supported SBC infrared sector preserves the dominant acoustic structure of the cosmic microwave background. The perturbative modifications remain concentrated in the infrared regime and do not generate large distortions at high multipoles.

This behavior is consistent with the interpretation of SBC as a weak infrared perturbative extension of the standard cosmological model rather than as a large modification of primordial acoustic physics.

4.3 Matter Power Spectrum Constraints

The matter power spectrum provides a direct probe of the evolution of cosmic structure and is therefore particularly sensitive to perturbative modifications of cosmological dynamics [15].

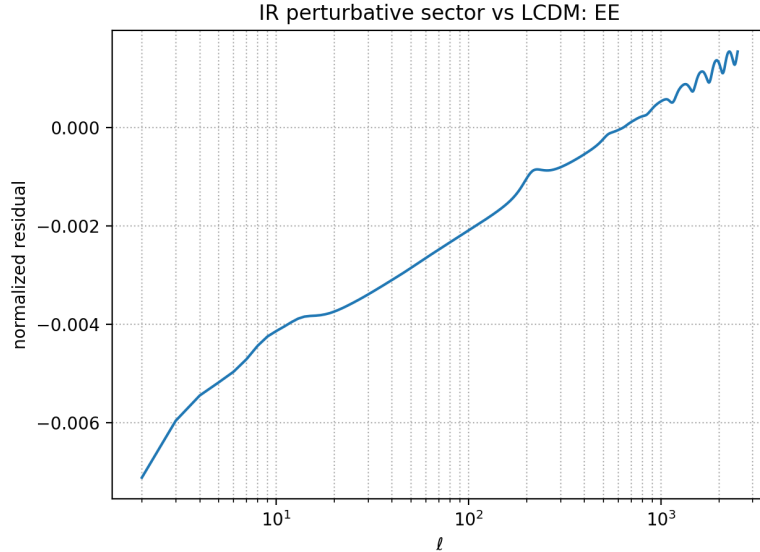


Fig. 3 Normalized residual diagnostic between the preferred SBC infrared sector and the reference Λ CDM model for the EE spectrum.

Unlike BAO and supernova observables, which primarily constrain the homogeneous expansion history, the matter power spectrum probes the growth of density fluctuations across a wide range of scales. Consequently, it constitutes one of the most relevant observables for testing the viability of infrared perturbative sectors.

Within the SBC framework, the infrared deformation introduced in Eq. (3) is expected to act predominantly at large scales while remaining sufficiently weak to preserve the overall consistency of the standard cosmological scenario.

The parameter-space exploration revealed the existence of a stable posterior-supported region in which matter-power modifications remain controlled and observationally acceptable. In particular, the preferred solutions identified by the combined analysis remain consistent with the posterior structure summarized in Table 2.

Within this region, the matter power spectrum exhibits small but non-vanishing deviations relative to the reference Λ CDM prediction. These deviations remain concentrated in the infrared regime and progressively decrease toward smaller scales.

An important outcome of the analysis is that the preferred infrared sector does not generate large-scale instabilities, excessive enhancement of clustering, or catastrophic suppression of structure formation. Instead, the deviations remain bounded and compatible with the observational constraints incorporated into the global validation framework.

The matter-power diagnostics therefore support the interpretation that the SBC infrared sector behaves as a weak perturbative correction rather than as a radical modification of the standard structure-formation scenario.

Taken together with the BAO and CMB analyses, these results indicate that the observationally viable SBC sector is characterized by a hierarchy in which background

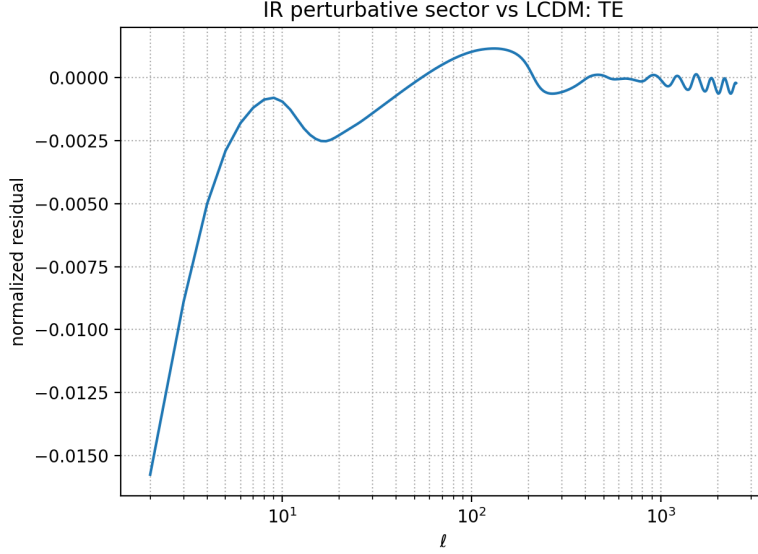


Fig. 4 Normalized residual diagnostic between the preferred SBC infrared sector and the reference Λ CDM model for the TE spectrum.

evolution remains effectively unchanged while perturbative observables retain a limited degree of infrared flexibility.

4.4 Growth Rate Constraints

The growth rate of cosmic structure provides a complementary test of cosmological models beyond background expansion observables.

In particular, measurements of the quantity

$$f\sigma_8(z), \quad (10)$$

combine information about the linear growth rate,

$$f(z) = \frac{d \ln D}{d \ln a}, \quad (11)$$

with the fluctuation amplitude σ_8 , making this observable especially sensitive to perturbative modifications of gravity and structure formation [15].

Because the SBC infrared sector is designed to preserve the homogeneous background evolution, growth-rate measurements constitute a critical test of whether perturbative deviations remain observationally acceptable.

The validation framework incorporated growth-rate consistency diagnostics together with matter-power indicators in order to evaluate the impact of the infrared deformation across the posterior-supported parameter region.

The analysis revealed that the observationally viable region identified through the combined constraints remains compatible with current growth-rate measurements. In

particular, the preferred sector does not generate the large departures from standard growth evolution that would typically accompany strong modifications of the cosmological background.

Instead, the resulting deviations remain moderate and concentrated within the infrared regime, consistent with the general hierarchy

$$|F_{\text{background}}| \ll |F_{\text{perturbative}}| \ll 1. \quad (12)$$

Independent growth diagnostics repeatedly favor parameter combinations that remain concentrated within the posterior-supported region summarized in Table 2, consistent with the regions identified through CMB, BAO, and matter-power analyses.

These results support the interpretation that the SBC infrared sector acts primarily through weak perturbative modifications of large-scale structure growth while leaving the standard cosmological background essentially unchanged.

Consequently, current growth-rate observations do not require the introduction of large background-level departures from Λ CDM and remain compatible with a background-preserving infrared perturbative scenario.

The growth diagnostics indicate that the preferred infrared sector modifies the perturbative evolution only at a moderate level and remains compatible with current large-scale-structure observations.

No evidence is found for runaway growth, large-scale instabilities, or strong departures from the standard growth history.

The resulting phenomenology therefore remains consistent with the interpretation of SBC as a weak infrared perturbative sector.

5 Global Posterior Structure

The independent observational diagnostics presented in the previous sections were combined into a global proxy likelihood in order to identify regions of parameter space that remain simultaneously compatible with all available constraints.

The combined analysis incorporates information from:

- Planck-inspired TT residual diagnostics,
- Planck-inspired EE residual diagnostics,
- Planck-inspired TE residual diagnostics,
- matter power-spectrum consistency metrics,
- growth-rate observables,
- Pantheon+ full-covariance distances,
- DESI DR2 baryon acoustic oscillation measurements.

Although the present framework does not constitute a full cosmological likelihood analysis, it provides an efficient phenomenological approach for identifying regions that remain compatible with multiple independent observational probes.

Posterior sampling of the resulting proxy likelihood was performed using the affine-invariant ensemble Markov Chain Monte Carlo algorithm implemented in the `emcee` package [21].

A notable feature of the resulting parameter space is the emergence of a stable infrared-safe sector that remains compatible with all observational diagnostics simultaneously.

Figure 5 presents the proxy-posterior reconstruction obtained from the MCMC sampling of the global proxy likelihood. The diagonal panels show the marginalized posterior distributions of the phenomenological parameters, while the off-diagonal panels display the corresponding joint posterior projections.

The posterior reconstruction reveals a localized probability concentration around weak deformation amplitudes and infrared scales of order 10^{-2} . The screening parameter exhibits a broader marginal distribution, indicating that current observations constrain the overall amplitude and characteristic infrared scale more strongly than the detailed shape of the screening profile.

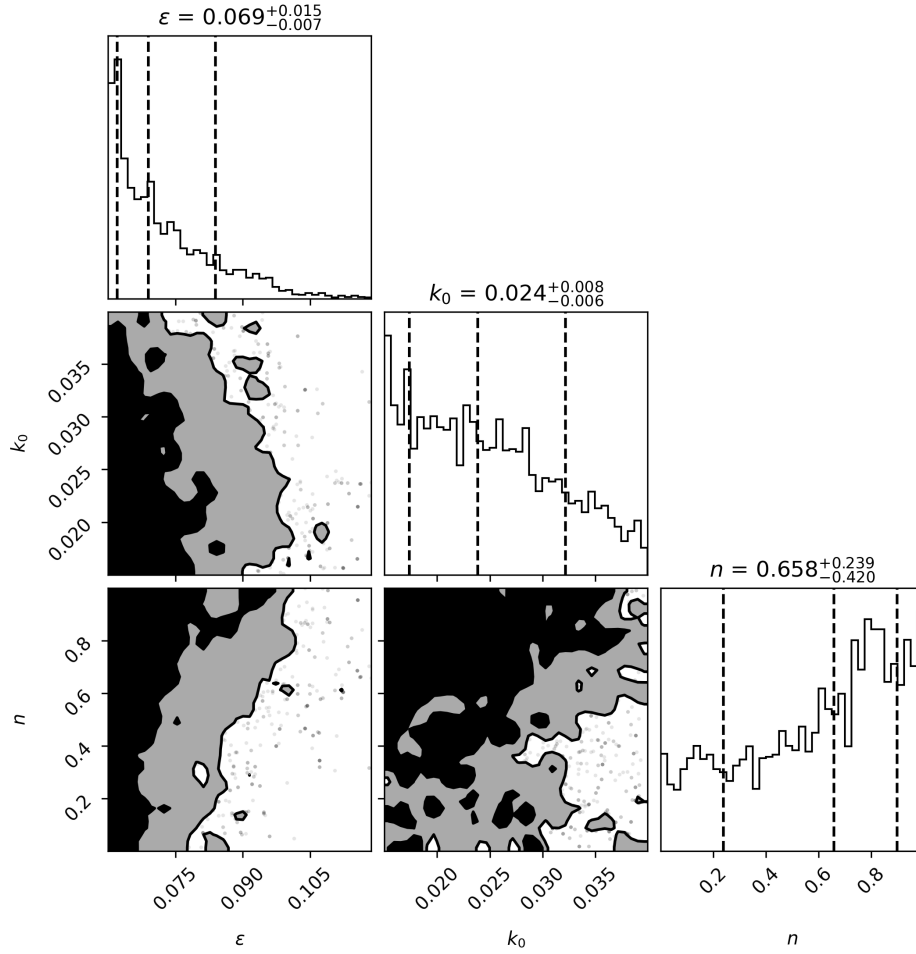


Fig. 5 Corner-plot representation of the proxy-posterior distribution obtained from MCMC sampling of the global proxy likelihood.

5.1 Proxy Posterior Constraints

Table 2 summarizes the proxy-posterior percentile estimates obtained from the global validation framework.

Table 2 Proxy-posterior percentile estimates for the SBC infrared-sector parameters. The intervals correspond to the 16th, 50th, and 84th percentiles of the proxy posterior chain.

Parameter	16th percentile	Median	84th percentile
ϵ	0.0620	0.0690	0.0839
k_0	0.0174	0.0239	0.0322
n	0.238	0.658	0.897

The proxy-posterior percentile estimates provide an indicative measure of parameter uncertainty within the phenomenological validation framework.

They should not be interpreted as full cosmological credible intervals, since the analysis does not sample the complete standard cosmological parameter space and does not employ the full Planck likelihood.

Nevertheless, the posterior structure exhibits a remarkable degree of stability across the independent observational diagnostics considered in this work.

The median posterior values are

$$\epsilon = 0.069, \tag{13}$$

$$k_0 = 0.0239, \tag{14}$$

$$n = 0.658, \tag{15}$$

with corresponding 16th–84th percentile ranges reported in Table 2.

A robust feature of the posterior reconstruction is the concentration of probability density around weak deformation amplitudes, infrared scales of order $k_0 \sim 10^{-2}$, and screening indices of order one, with posterior support extending toward the minimal exponential regime.

The marginal distribution of n is broader than the corresponding distributions of ϵ and k_0 , indicating that the screening profile remains less tightly constrained within the present proxy-likelihood framework.

This behavior suggests that the observational constraints primarily determine the characteristic amplitude and infrared scale of the effective deformation, while allowing greater flexibility in the detailed screening profile.

Taken together, the posterior reconstruction supports the existence of an observationally viable infrared-safe perturbative sector that preserves the standard Λ CDM background evolution while allowing weak perturbative deviations compatible with current cosmological observations.

The resulting posterior region should be interpreted as a phenomenological reference point within the adopted parametrization rather than as a fundamental prediction of the SBC framework.

Future theoretical developments may determine whether similar infrared responses emerge from more fundamental causal-memory constructions or from broader classes of effective descriptions.

6 Memory-Kernel Interpretation

The observational analysis presented in the previous sections identifies an infrared-safe perturbative sector supported by the global proxy-posterior reconstruction summarized in Table 2.

Although the present work focuses primarily on the phenomenological characterization of this sector, it is natural to ask whether the observed infrared behavior admits a physical interpretation within the broader SBC framework.

In SBC, observable dynamics need not constitute a complete description of the total causal structure. Instead, the observable sector may represent only a restricted projection of a larger causal system. Similar ideas arise in statistical coarse-graining approaches, projection-operator methods, and reduced descriptions of complex dynamical systems, where unresolved degrees of freedom can generate effective memory terms in the observable dynamics [12–14].

The purpose of this section is therefore not to derive a complete fundamental theory, but rather to outline a possible interpretation connecting the observationally viable infrared sector with an emergent causal-memory description.

6.1 Observable–Hidden Decomposition

A central idea within SBC is that the observable state of a physical system may not exhaust the complete causal information available in the underlying dynamics.

We therefore consider a schematic decomposition of the total state,

$$\Psi_{\text{tot}} = \Psi_{\text{obs}} + \Psi_{\text{hid}}, \quad (16)$$

where Ψ_{obs} denotes the observable sector and Ψ_{hid} represents degrees of freedom that are not directly accessible within the observational window.

The total causal dynamics may then be written schematically as

$$\frac{d\Psi_{\text{obs}}}{dt} = \mathcal{L}_{\text{obs}}\Psi_{\text{obs}} + \mathcal{C}\Psi_{\text{hid}}, \quad (17)$$

$$\frac{d\Psi_{\text{hid}}}{dt} = \mathcal{L}_{\text{hid}}\Psi_{\text{hid}} + \mathcal{D}\Psi_{\text{obs}}, \quad (18)$$

where $\mathcal{L}_{\text{obs}}$ and $\mathcal{L}_{\text{hid}}$ denote the intrinsic dynamics of the two sectors and $\mathcal{C}$ and $\mathcal{D}$ describe their mutual coupling.

Because the hidden sector is not directly observable, its influence can only be inferred indirectly through its effect on the evolution of the observable variables.

This observable–hidden decomposition is conceptually similar to the partitioning employed in projection-operator approaches, where unresolved variables are integrated out in order to construct an effective description of the accessible sector [12, 13].

6.2 Emergent Memory

If the hidden sector cannot be directly resolved within the observational window, its degrees of freedom may be formally eliminated from the effective description.

In generalized projection frameworks, the elimination of unresolved variables naturally generates nonlocal evolution equations containing memory kernels that encode the residual influence of the hidden sector [12, 13].

Motivated by this perspective, the observable-sector evolution may be represented schematically as

$$\frac{d\Psi_{\text{obs}}}{dt} = \mathcal{L}_{\text{eff}}\Psi_{\text{obs}} + \int_{-\infty}^t K(t, t')\Psi_{\text{obs}}(t') dt', \quad (19)$$

where the kernel $K(t, t')$ summarizes the accumulated influence of the eliminated degrees of freedom.

In this interpretation, memory is not introduced as an independent postulate. Rather, it emerges naturally from incomplete causal accessibility and from the coarse-grained description of a larger dynamical system.

The resulting conceptual chain may be summarized schematically as

$$\text{finite observational access} \rightarrow \text{hidden sector} \rightarrow \text{effective memory}. \quad (20)$$

Consequently, memory effects appear as an effective description of information that remains inaccessible within the observable projection.

It is important to emphasize that the present work does not derive the phenomenological response function of Eq. (3) from the kernel $K(t, t')$. The equations above are intended only as a conceptual illustration of how infrared effective responses may emerge from hidden-sector dynamics within the broader SBC framework.

A complete derivation connecting the phenomenological infrared sector to an explicit memory kernel remains an open problem and is left for future investigation.

7 Discussion and Conceptual Interpretation

The previous section outlined a possible interpretation of the observationally viable infrared sector in terms of emergent memory effects associated with incomplete causal accessibility.

The purpose of the present section is to discuss the broader conceptual implications of this interpretation and to clarify its status within the SBC framework.

At the phenomenological level, the observational analysis identifies a stable infrared-safe sector characterized by weak perturbative deviations and negligible background deformation. However, the existence of such a sector does not by itself establish a unique underlying physical mechanism.

The SBC framework proposes that the observed infrared behavior may arise from an effective observable–hidden decomposition of the total causal dynamics. Within this interpretation, memory effects emerge naturally from the elimination of inaccessible degrees of freedom, generating an effective nonlocal description at the observable level.

The following subsections discuss the simplest causal-memory realization compatible with the observational results and examine the extent to which the present phenomenological findings may support the broader SBC conceptual framework.

7.1 Minimal Causal Kernel

The emergence of an effective memory term naturally raises the question of which kernel structure provides the simplest stable description of the hidden-sector influence.

A physically acceptable kernel must satisfy three basic requirements.

First, it must preserve causal ordering, implying that future states cannot influence past observable configurations.

Second, the kernel must remain normalizable in order to avoid divergent long-term memory accumulation.

Third, the resulting evolution must remain dynamically stable under repeated application of the memory operator.

The simplest kernel satisfying these conditions is an exponentially screened causal response,

$$K(t, t') = K_0 e^{-\lambda(t-t')} \Theta(t - t'), \quad (21)$$

where Θ denotes the Heaviside causal-support function.

The exponential factor suppresses increasingly distant contributions from the past, while the Heaviside term guarantees causal ordering.

This kernel therefore represents the minimal stable memory response compatible with the observable–hidden decomposition introduced in the previous section.

When mapped schematically onto cosmological variables, the corresponding leading infrared response takes the approximate form

$$F_{\text{SBC}}(z) \simeq -\epsilon e^{-k_0(1+z)}. \quad (22)$$

This expression corresponds to the simplest exponential realization of the more general phenomenological parametrization introduced in Eq. (3),

$$F_{\text{SBC}}(z) = -\epsilon \exp [-(k_0(1+z))^n]. \quad (23)$$

The current proxy-posterior reconstruction presented in Section 5 does not uniquely select the minimal exponential realization.

The posterior median of the screening parameter is

$$n = 0.658, \quad (24)$$

with a 16th–84th percentile interval

$$0.238 < n < 0.897. \quad (25)$$

Consequently, the present analysis does not provide evidence that the minimal exponential regime $n = 1$ is uniquely preferred by the observational data.

Instead, the observationally viable region includes a broader family of screening profiles, of which the exponential case represents one particularly simple limiting realization.

Nevertheless, the fact that part of the posterior support extends into the vicinity of $n \sim 1$ suggests that the observationally viable infrared sector remains compatible with a simple exponentially screened causal-memory response.

Within the SBC interpretation, this compatibility is suggestive of a possible connection between the observed infrared phenomenology and emergent causal-memory dynamics. However, no unique causal-memory kernel can be inferred from the present phenomenological analysis.

This interpretation should therefore be regarded as suggestive rather than demonstrative, since the present work does not derive Eq. (3) from an explicit microscopic kernel.

7.2 Foundational Status of SBC

The observational results presented in this work establish the existence of an infrared-safe perturbative sector that remains compatible with current cosmological observations while preserving the standard Λ CDM background expansion history.

Within the phenomenological framework explored here, the proxy-posterior reconstruction summarized in Table 2 constitutes the principal empirical result of the present study.

However, the observational identification of a viable infrared regime does not by itself determine the unique physical origin of the corresponding phenomenology.

The SBC framework proposes one possible interpretation in which the observed infrared behavior emerges from an effective observable–hidden decomposition of the total causal dynamics. Under this interpretation, memory effects arise naturally through the elimination of inaccessible degrees of freedom, generating an effective nonlocal description at the observable level.

At present, this interpretation should be regarded as a conceptual framework rather than a fully established fundamental theory.

The observational analysis demonstrates that the phenomenological SBC infrared sector is observationally viable. It does not yet demonstrate that the proposed causal-memory interpretation is unique, nor does it establish that the minimal exponential kernel discussed above is the only possible realization.

Consequently, the present work should be viewed as identifying a phenomenologically successful infrared sector together with a physically motivated interpretative framework that remains open to further theoretical development.

Future work should determine whether the preferred infrared response can be derived from a more fundamental formulation of SBC and whether exponentially screened kernels emerge naturally from deeper causal principles.

In this sense, the current results may be interpreted as an observational foundation upon which a more complete SBC theoretical structure could eventually be constructed.

An important open question concerns the extent to which projection-based approaches such as Mori–Zwanzig theory restrict the admissible class of infrared responses.

If future developments show that observable–hidden decompositions naturally favor restricted families of exponentially screened or finite-memory kernels, the

observationally viable region identified in the present work could provide a useful phenomenological target for such constructions.

The present analysis does not establish this connection, but it helps define a concrete observational regime against which future causal-memory models may be compared.

The observationally viable region identified in the present work should therefore be interpreted as a phenomenological target within the specific infrared parametrization adopted here. Future causal-memory constructions may reproduce this regime through different kernel realizations, provided they remain compatible with the observational constraints established by the present analysis.

More generally, the present results provide a reproducible observational benchmark for future causal-memory formulations. An important open question is whether projection-based hidden-sector dynamics, such as those described by Mori–Zwanzig approaches, naturally favor kernel families capable of reproducing the infrared regime identified here, or whether more general memory constructions are required. The answer to this question lies beyond the scope of the present work and constitutes a natural direction for future investigation.

8 Status and Limitations

The results presented in this work should be interpreted within the scope of the phenomenological framework employed in the analysis.

A principal strength of the study is the combination of multiple independent cosmological probes, including CMB diagnostics, baryon acoustic oscillation measurements, Type Ia supernova distances, growth-rate indicators, and matter power-spectrum consistency tests.

The analysis demonstrates that an observationally viable infrared-safe sector exists within the explored SBC parameter space and that this sector remains compatible with current cosmological observations while preserving the standard Λ CDM background expansion history.

At the same time, several limitations should be emphasized.

First, the present analysis relies on compressed and proxy-likelihood constructions rather than on the full Planck likelihood pipeline. Although the compressed diagnostics preserve the dominant observational information, they do not replace a complete cosmological parameter-estimation framework.

Second, the posterior reconstruction presented here is derived from a proxy-likelihood framework and does not sample the full standard cosmological parameter space simultaneously with the SBC sector. A complete Markov Chain Monte Carlo analysis jointly exploring both sectors remains an important objective for future work.

Third, the present study does not include Bayesian evidence calculations. Consequently, the results establish observational viability rather than formal model selection relative to Λ CDM.

Fourth, the theoretical interpretation proposed within the SBC framework remains incomplete. While the observable–hidden decomposition and emergent-memory picture provide a physically motivated explanation for the preferred infrared behavior,

the present work does not derive the phenomenological sector from a fully developed fundamental theory.

An additional limitation concerns the relationship between the phenomenological infrared sector studied here and the broader SBC framework.

While the effective response function investigated in this work is motivated by SBC concepts, no first-principles derivation from a complete SBC dynamical theory has yet been established.

Moreover, the observationally preferred parameter region identified here is specific to the effective parametrization adopted in Eq. (3).

Whether similar observational behavior would emerge within broader classes of causal-memory kernels remains an open question.

Consequently, the present results should be interpreted as an observational assessment of an SBC-inspired phenomenological sector rather than as a direct test of a fully formulated SBC theory.

Another limitation concerns the current parameter constraints themselves. While the amplitude parameter ϵ and the characteristic infrared scale k_0 exhibit relatively localized posterior distributions, the screening index n remains more weakly constrained within the present proxy-likelihood framework. Additional observational information and full cosmological likelihood analyses will be required to determine the detailed structure of the screening behavior more robustly.

Therefore, the principal conclusion of the present analysis is not that SBC has been established as a complete cosmological theory, but rather that a well-defined infrared perturbative sector exists that remains observationally viable and admits a natural interpretation within the broader SBC conceptual framework.

Future work should focus on four main directions:

- full cosmological likelihood analyses,
- Bayesian model-comparison studies,
- improved posterior reconstruction using complete cosmological parameter sampling,
- investigation of possible fundamental realizations of the infrared sector within the broader SBC framework.

These developments will be necessary to determine whether the phenomenological success of the present infrared sector reflects a deeper causal structure or merely represents one effective description among several possible alternatives.

Future work will investigate the preferred infrared sector using full Bayesian parameter estimation and cosmological MCMC analyses.

9 Conclusions

The objective of the present work was to investigate whether an effective infrared perturbative sector motivated by the SBC framework could remain compatible with current cosmological observations while preserving the successful background evolution of the standard Λ CDM model.

To address this question, we combined multiple independent observational diagnostics, including Planck-inspired CMB residual analyses, baryon acoustic oscillation

measurements, Pantheon+ Type Ia supernova distances, growth-rate indicators, matter power-spectrum consistency tests, and a global proxy-posterior reconstruction.

The combined analysis reveals the existence of an observationally viable infrared-safe sector summarized in Table 2. The proxy-posterior structure favors weak deformation amplitudes, infrared scales of order $k_0 \sim 10^{-2}$, and moderately screened infrared responses, with posterior support extending toward the minimal exponential regime, while indicating broader uncertainty in the screening profile than in the deformation amplitude or infrared scale. Importantly, the present analysis establishes observational viability rather than observational preference relative to Λ CDM.

No Bayesian evidence calculation is performed, and therefore the results should not be interpreted as demonstrating a statistically preferred alternative to the standard cosmological model.

A central result of the study is that the preferred sector preserves the standard Λ CDM background expansion history to a high degree of accuracy. The combined BAO and Pantheon+ constraints strongly favor

$$\alpha_{\text{bg}} \simeq 0, \quad (26)$$

indicating that viable SBC realizations cannot be interpreted as large background-level modifications of the standard cosmological model.

Instead, the preferred solutions correspond to weak perturbative infrared corrections that leave the dominant acoustic structure of the cosmic microwave background essentially unchanged while allowing controlled deviations in observables associated with the growth of large-scale structure.

An important outcome of the analysis is the remarkable stability of the preferred infrared sector across independent observational probes. Independent observational probes consistently identify compatible regions of parameter space, suggesting that the infrared-safe sector is not driven by a single dataset but rather emerges as a common feature of the combined phenomenological framework.

Beyond its observational implications, the present work outlines a possible interpretation within the broader SBC program. In this perspective, observable infrared effects may arise from an effective observable–hidden decomposition of the underlying causal dynamics, leading naturally to nonlocal memory kernels at the observable level.

Within this interpretation, the posterior-supported infrared sector remains compatible with simple exponentially screened causal-memory responses such as

$$K(t, t') = K_0 e^{-\lambda(t-t')} \Theta(t - t'). \quad (27)$$

The current proxy-posterior reconstruction does not uniquely select this realization, although part of the posterior support extends into the vicinity of the minimal exponential regime. Although no first-principles derivation is established in the present work, this interpretation provides a physically motivated framework linking the observed infrared phenomenology to a broader causal-memory perspective.

At the same time, the results presented here should not be interpreted as establishing SBC as a complete cosmological theory. Rather, they demonstrate that a well-defined infrared perturbative sector exists that remains compatible with current observations and admits a coherent phenomenological interpretation.

Several important limitations remain. The present analysis employs proxy likelihoods rather than the full Planck likelihood, does not perform a complete cosmological MCMC exploration of the standard parameter space, and does not yet provide Bayesian model comparison relative to Λ CDM. Consequently, the reported parameter intervals should be interpreted as proxy-posterior estimates within the phenomenological framework adopted here.

Future work should focus on full cosmological likelihood analyses, Bayesian evidence calculations, cosmological MCMC parameter estimation, reproducible public implementations of the SBC framework, and the investigation of possible fundamental realizations of the infrared sector within a more complete causal description.

In summary, the principal outcome of this work is the identification of an observationally viable infrared perturbative sector that preserves the standard cosmological background while allowing controlled large-scale perturbative effects. This result provides a reproducible observational benchmark for future investigations of infrared cosmological phenomenology and its possible connection to emergent causal-memory descriptions.

Whether such descriptions ultimately arise from a fundamental SBC formulation or from broader classes of effective infrared theories remains an open question for future work.

Reproducibility Statement

The numerical implementation used in this work is based on publicly available cosmological datasets together with custom validation and proxy-likelihood analysis scripts.

The complete reproducibility package associated with this work, including MCMC chains, posterior summaries, diagnostic figures, analysis scripts, and documentation, is publicly available through the SBC-IR-Observational-Viability repository and permanently archived on Zenodo:

<https://doi.org/10.5281/zenodo.20720308>

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Declarations

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Data availability. All observational datasets employed in this work are publicly available from their respective collaborations.

Code availability. The numerical implementation, validation scripts, posterior chains, figures, and reproducibility material are publicly available through the SBC-IR-Observational-Viability repository and the associated Zenodo archive (DOI: 10.5281/zenodo.20720308).

Author contributions. Carlos Hernan Quintero B. developed the theoretical framework, observational analysis strategy, numerical validation pipeline, and manuscript preparation. Emili Quintero Castillo contributed to conceptual development, interpretation of results, and manuscript review.

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Data and Code Availability

All scripts, parameter grids, posterior chains, figures, and reproducibility materials used in this work are publicly available through the SBC-IR-Observational-Viability repository.

GitHub repository:

<https://github.com/causalidad/SBC-IR-Observational-Viability>

Permanent Zenodo archive:

<https://doi.org/10.5281/zenodo.20720308>

The archived repository contains the complete reproducibility package associated with the observational benchmark presented in this work.

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